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COLLEGE OF CREATIVE INNOVATION

11

Programme

Master of Software Engineering – 180 credits

Work In Progress Report

Assesement 2

Project Title:

Blockchain-Based QR Code Verification System

Submitted by:

Perisha Dharel (270629406)

7

Course Code:

MSE907 Industry Based Capstone Research Project

Tutor Name:

Dr. Usman Ghani

Date of Submission:

15 March 2026

11

Yoobee Colleges of Creative Innovation

School of Technology Auckland,

New Zealand

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1 Introduction

Agreement checking is becoming very important with the problem of credential fraud and manipulation in schools, businesses, and certifying officials. Conventional verification mechanisms based on centralized databases can be attacked by insiders, modified without authorization and be vulnerable to single points of failure. The immutable, decentralized storage of data provided by blockchain would be a secure method of verifying the digital credentials by anchoring the hash of the cryptographic documents to a distributed ledger [1–3].

Although there are many blockchain-based solutions, most of them are based on expensive public networks, and the issues of privacy compliance remain a concern, as well as they do not fit small institutions with scarce technical resources. This work in progress report informs the latest position of research and development and preliminary discussion of ethics and cultural issues of the proposed Blockchain-Based QR Code Verification System.

2 Background and Problem Statement.

The main weakness of centralized document verification platforms is that data may be tampered with and not be transparent. Earlier studies revealed that blockchain can help alleviate the problems by storing cryptographic fingerprints of documents to be verified without trust [3–5]. Nevertheless, numerous blockchain systems have sensitive information stored directly on-chain, which is in contrast to the privacy laws such as the GDPR and the NZ Privacy Act 2020 [4]. Furthermore, the reliance on public network results in the high cost of gas and technical overhead that are not suitable to small educational or certification providers[2,3].

Therefore, the essence of the issue that this research examines is:

“But how can a privacy conscious, lightweight and cost efficient blockchain validation system based on using SHA-256 hash and QR codes be able to verify documents with high security without revealing sensitive information and without using expensive infrastructure resources?”

3. Objectives of the Project

The main goals that will be used to guide this WIP are:

- In order to further develop the literature review with the latest research, to expand the knowledge on blockchain verification and QR integration.
- To narrow down research questions and hypotheses with the emergent evidence and implementation issues.
- To design and implement the prototype of the system based on Django, Solidity smart contracts, and Web3.py.
- To examine initial performance, security, and usability factors.

In order to detect ethical, legal, and cultural issues at a tender stage into the project lifecycle.

4. Literature Review

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The immutable character of the blockchain technology has been commonly accepted as a viable solution to the issue of a secure document verification since it is decentralised. According to the previous research, document integrity is guaranteed by storing cryptographic hashes on-chain, preserving the privacy of data by keeping original files off-chain [4,6]. These methods help decrease the forgery of documents and increase the confidence in the verification systems.

Nonetheless, there are a number of limitations which are shown in existing literature. Most of the solutions in blockchain are theoretical and not system-implemented, especially in the institutional context. It is also stated that scalability and transaction latency are the current problems to deal with, particularly, when one has to process large numbers of verification requests within the system [5].

Integration of the QR codes has also been suggested in order to enhance the accessibility and verification ease. Studies suggest that systems with QR have fast verification without specific technical expertise, and thus they can be used by different users (Valadares et al., 2023). Regardless of these benefits, security threats like QR code copying and inadequate connection with blockchain verification logic are not thoroughly researched.

Besides, the majority of the studies focus on the technical performance metrics with a little consideration of usability and user experience evaluation. There are not many systems which employ mixed-method evaluation solutions that involve performance testing and structured tools of usability testing. This gap shows the necessity of solution that is secure, performance-wise and accessible to users.

The proposed project will fill these gaps by introducing a blockchain-based QR code certificate verification system, which incorporates smart contracts, focuses on usability, and uses quantitative and qualitative assessment tools.

Although the prior literature is concerned with the aspects of performance and hash integrity [4,6], the recent research (e.g., [5,7]) emphasizes the necessity of incorporating usability assessment along with the concept of security. This project also fills these gaps by integrating blockchain hash verification and QR codes with the usability testing to present a safe and convenient system.

5. Research Gaps or Challenges.

The system also employs lightweight local blockchain instead of the public networks to mitigate high cost and complexity. The compliance of privacy is met by the storage of hashes of SHA-256 only on-chain. The incorporation of QR codes will increase the usability and discourage cloning since it directly connects with transactions in smart contracts. Modular design reduces the problem of scalability in that, system components can be tested independently.

From the literature:

- **Cost and Complexity:** The majority of blockchain systems rely on the use of public networks (e.g., Ethereum), which have extremely high transaction costs that small institutional budgets cannot afford [2,3,8].
- **Privacy Compliance:** The storage of personal or sensitive information on-chain infringes privacy principles of NZ Privacy Act 2020 and GDPR [4].
- **Integrated Lightweight Design:** There are not many platforms that merge blockchain hash, the use of QR codes, and privacy-focused storage into a working prototype [2,9,10].

- **Usability Problems:** QR codes that are not cryptographically linked can be cloned to reduce their trustworthiness.

Such deficiencies guide the basic structure of the suggested system local blockchain, hash-only on-chain storage, and QR based on transaction IDs.

6. Research Questions/ Hypotheses.

Refined Research Questions:

- RQ1: Does a local blockchain system based on SHA 256 hashing reliably identify altered documents?
- RQ2: Does privacy compliance with storing cryptographic hashes only on-chain?
- RQ3: Will the integration of QR codes increase the usability of non technical verifiers?
- RQ4: Does a lightweight blockchain verification solution make sense and cost effectively to small institutions?

Hypotheses:

- H1: SHA -256 hashing guarantees proper tampering.
- H2: Hash-only on-chain storage: GDPR and NZ privacy standards.
- H3: Local blockchain systems have low latency verification.
- H4: QR code integration leads to higher usability than manual verification.

7. Methodology / Approach

The project is based on a design-science and experimental approach in order to work out a lightweight and privacy-saving system of document verification. The methodology will involve a combination of system design, prototype development, performance experiments, and usability testing as the solution to the existing gap identified of generating the lightweight blockchain verification of small institutions.

- **Prototype Development:** The implementation of the system will be done with Django as the backend APIs, Web3.py as the blockchain interactions, and smart contracts in a local blockchain on Ganache. SHA 256 hashing provides integrity in documents. The results of the experiment will confirm the RQ1-RQ4: hash production and transaction time will prove the reliability of the system (RQ1 and RQ3), SUS and usability as a task will determine the efficiency of the QR code integration (RQ3 and RQ4). Agile principles are applied in development such that sprints are done weekly and integrated continuously with iterative integration to have incremental delivery and constant improvement.
- **Experimental Evaluation:** Performance measures such as hash computation time, transaction latency and QR code verification performance rate will be gathered and analyzed in a quantitative format.
- **Usability Testing:** The system usability scale (SUS) survey and the task-based testing with people who are not technical will be used to measure the user experience and accessibility.
- **Ethical Assessment:** A qualitative analysis will ensure the adherence to privacy standards (NZ Privacy Act 2020, GDPR) and overcome the risks of system design biases [4,11].

8. Tools and Technologies Used

Django and Web3.py were chosen because they are easily integrated with Python and have REST API support and can be used with Ganache to simulate a local blockchain. SHA-256 has been selected as a cryptographic hash computation as it provides sufficient security and can be used in small institutional applications.

- **Programming Languages:** Python (Django, Web3.py), Solidity.
- **Ethereium blockchain:** Ganache (local blockchain ethereum simulation)
- **Database:** MySQL (off-chain metadata of documents)
- **Frontend:** JavaScript, Bootstrap, CSS and HTML.
- **Version Control:** Git and GitHub
- **Hashing:** SHA 256 (cryptographic document fingerprinting)

- **Testing:** PyTest which is used in unit testing, Selenium which is used in integration testing.
- **Design Tools:** Lucidchart (as an architecture diagram), draw.io (as a flowchart).

Such technologies facilitate a lightweight and scalable and secure system and can be used in small-scale institutional settings [3,5].

9. System architecture or overview design

The system architecture is modular with separation of concerns, and it promotes the Agile iterative development. All its layers (frontend, backend, blockchain) can be developed, tested, and refined independently, which encourages maintainability and constant integration. The system follows three layer architecture:

- **Frontend Layer:** Document uploading interface, QR code scanning interface, and verification result display interface.
- **Backend Layer:** Django REST APIs will handle uploading of documents, creating SHA -256 hashes, communicating with smart contracts and recording verification of findings.
- **Blockchain Layer:** Ganache blockchain stores hash of documents of smart contracts only which cannot be altered and keeps user data off-chain as privacy compliance.

Diagram Overview:

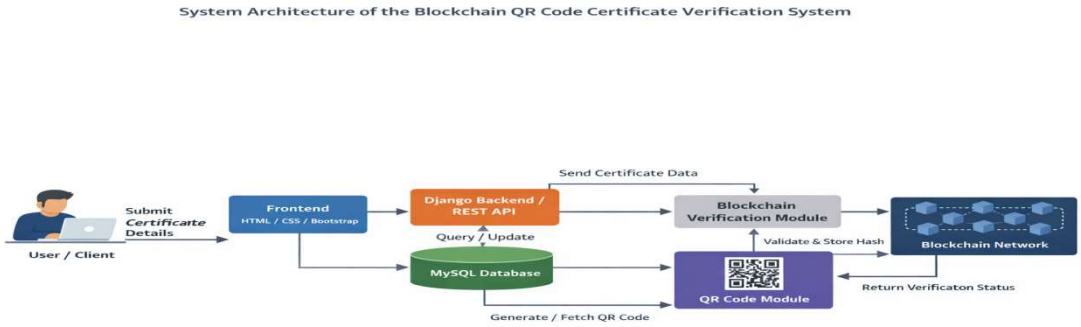

Figure 1: System Architecture or Overview

The architecture provides modularity, scalability, and separation of concerns that easily support maintenance and testing as well as complies with software engineering principles [10].

10. Implementation Plan (Week 7 -Week 13)

The implementation phase is done in an incremental and modular implementation based on the system architecture.

- **Week 7:** Environment Setup and Architecture : Alignment Development tools such as Django, Python virtual environment and Ganache are set up. Django rest framework is incorporated and Web3.py linkage between blockchain and back-end is made.

- **Week 8:** Smart Contract Development: Solidity smart contracts will be developed to store and retrieve document hashes. The deployment of contracts is done in Ganache, and unit testing is performed to ensure the immutability and accuracy [3](Bhaskar & Reddy, 2022).
- **Week 9:** Backend Development: Backend APIs Backend APIs are deployed to process document uploads, create SHA-256 hashes, and communicate with smart contracts. Authentication controls are brought to limit access by unauthorized access.

- **Week 10:** QR Code Integration: QR codes are created with reference to blockchain transactions. The system allows the QR scanning, which can be used to retrieve and verify the hashes of corresponding documents [10].

- **Week 11:** Frontend Development: Frontend development is completed with the Django templates that offer users the opportunity to upload files, scan QR codes, and obtain the results of verification.

- **Week 12:** System Integration and Testing: The entire modules are end to end integrated and tested. The system is tested against the document tampering situations to capture the scene.

- **Week 13:** Evaluation Preparation: Primary security, performance and usability evaluations are done. System Usability Scale (SUS) is ready to be tested by the user [12].

11. Assessment and Evaluation Plan

The evaluation project method is mixed method:

- **Quantitative Testing:** RQ1 and RQ3 will be directly answered using performance indicators including the time taken to compute a hash using the SHA-256 hash-function and the latency of document verification, which will demonstrate the reliability of the hash-computation and low-latency document-verification functions of the hash algorithm. RQ4 will be met by success rates of QR scans on documents of different sizes, proving that they have increased usability compared to manual verification.
- **Qualitative Testing:** RQ3 and RQ4 will be tested directly using the Survey, the System Usability Scale (SUS), as well as using the method of task-based user testing with non-technical participants, which will help to understand whether the integration of QR codes can enhance the user experience and accessibility.
- **Security Testing:** RQ1 and RQ2 will be confirmed by controlled testing of hash collision, manipulation of smart contracts and QR code cloning, which will ensure the system is reliable in detecting any tampering and adhering to privacy requirements.

12. Ethical, Legal, and Cultural Concerns.

- **Ethical:** The privacy of the users is ensured because only the SHA-256 hashes are sent to the blockchain, whereas documents are stored off-chain. The user rights are explicitly given when uploading the data and smart contracts are strictly checked to avoid hacking and intrusion.
- **Legal:** The system is in line with the New Zealand Privacy Act 2020 and GDPR and has defined data retention policy of off chain storage.
- **Cultural:** The software will facilitate multilingual interfaces and easy design to suit different and non-technical users. The considerations of deployment involve usability within institutions that have a low technical infrastructure.

Participants with various technical experience will be included in usability testing to reduce the effect of bias and make sure that the system is accessible and inclusive.

13. Timeline and Project Plan

The Blockchain-Based QR Code Verification System has a well-organized 15-week plan of implementation in Agile, which includes the proposal development and system implementation phases. The plan has systematic movement between the research and design and implementation, testing and final evaluation. Individual system components were delivered by sprinting every week in order to provide continuous integration, testing, and refinement. This design allows flexibility, pre-election of technical issues, and compatibility with the goals of the research.

The plan of the project is structured in the following way:

- **Weeks 1-6:** Development of research proposal, which will include literature review, gap analysis, research design, and submission Assessment 1.
- **Weeks 7-13:** Core system design, including environment preparation, implementation of smart contracts, backend development, blockchain integration, functionality of QR codes, frontend design and system integration.
- **Weeks 14 -15:** Security testing, performance evaluation, deployment, documentation and final presentation.

This gradual schedule will guarantee the delivery on time without compromising on academic excellence and technical competence.

14. Status and Completed Deliverables (Work in Progress)

Weeks 7-13 have been effectively completed as of the present level of the project. It has attained the following deliverables:

- The development environment was set up with the help of Python, Django, Ganache, Solidity, and Web3.py.
- SHA-256 hash detection algorithm used in the generation of secure document fingerprints.
- Smart contracts were built and run on local Ganache blockchain to store hash of documents.
- Django was used to connect with the blockchain with Web3.py.
- QR code and scanning features associated with blockchain identifiers of transactions.
- Frontend interface that is user friendly and python based on Django templates, HTML, and CSS.
- Complete integration of the system with end to end verification of documents.
- Primary functionality testing is done to ensure accuracy and reliability of the system.

These completed components affirm the fact that the proposed system architecture is technically viable and meets the stipulated research objectives.

15. Timeline of Pending Deliverables

The rest of the activities will take place in Weeks 14 and 15 and will be based on evaluation, optimisation, and final documentation:

Week 14 (Ongoing):

- Security testing to determine the resistance to hash collisions, QR code cloning and unauthorised access.
- Performance measurement in terms of latency, time of hash generation and speed of QR verification.
- Small-scale optimisation and cloud deployment of institutional use.

Week 15 (Pending):

- Last system testing and verification of research hypotheses.
- Testing results: Bug fixing and system refinement.
- Final project report and preparation of presentation material.

These deliverables, which are pending, will give an empirical evidence that the proposed Blockchain-Based QR Code Verification System is effective, secure, usable, and feasible.

References

1. Abidin, M. F. & Munggaran, L. C. Blockchain-Based Digital Document Verification Using SHA-256 on the Internet Computer Protocol (ICP). *J. Electr. Electron. Inf. Commun. Technol.* **7**, 64–68.
2. Sawant, D., Potdukhe, S., More, V., Apsingekar, P., & Mahadik, A. Digital document verification using blockchain. *IJRASET Journal for Research in Applied Science and Engineering Technology* https://doi.org/10.22214/ijraset.2025.68159 doi:https://doi.org/10.22214/ijraset.2025.68159.
3. S. Bhaskar & K. Reddy. Bhaskar, S., & Reddy, K. (2022). Tamper-proof academic certificate authentication using Ethereum blockchain. IEEE Access, 10, 25634–25645. https://doi.org/10.1109/ACCESS.2022.3151527. *IEEe Access* **10**, 25634–25645.
4. Haque, A. B., Islam, A. N., Hyrynsalmi, S., Naqvi, B. & Smolander, K. GDPR compliant blockchains–a systematic literature review. *Ieee Access* **9**, 50593–50606 (2021).
5. Noorhizama, N. K., Abdullah, Z., Kasim, S., Hamid, I. R. A. & Isa, M. A. M. Verification of ph. d. certificate using qr code on blockchain ethereum. *JOIV Int. J. Inform. Vis.* **7**, 716–725 (2023).

6. Ayush Mishra, Sakshi Mehta, Bhavya Oza, Sumit Kumar, Hemant Kasturiwale. Blockchain-based decentralized document verification and its applications. *Journal of Information Systems Engineering and Management* **10**, (2025).

7. Singh, C., Thakkar, R. & Warraich, J. IAM Identity Access Management—Importance in Maintaining Security Systems within Organizations. *Eur. J. Eng. Technol. Res.* **8**, 30–38 (2023).

8. AlShamsi, M., Al-Emran, M. & Shaalan, K. A systematic review on blockchain adoption. *Appl. Sci.* **12**, 4245 (2022).

9. Rahman, T., Mouno, S. I., Raatul, A. M., Al Azad, A. K. & Mansoor, N. Verifi-chain: a credentials verifier using blockchain and IPFS. in *International Conference on Information, Communication and Computing Technology* 361–371 (Springer, 2023).

10. Gangwar, S. Blockchain-based Authentication and Verification System for Academic Certificate using QR Code and Decentralized Applications. *Int. J. Comput. Appl.* **975**, 8887.

11. Valadares, D. C. G., Perkusich, A., Martins, A. F., Alshawki, M. B. & Seline, C. Privacy-preserving blockchain technologies. *Sensors* **23**, 7172 (2023).

12. Cardenas-Quispe, M. A. & Pacheco, A. Blockchain ensuring academic integrity with a degree verification prototype. *Sci. Rep.* **15**, 9281 (2025).

Screenshots

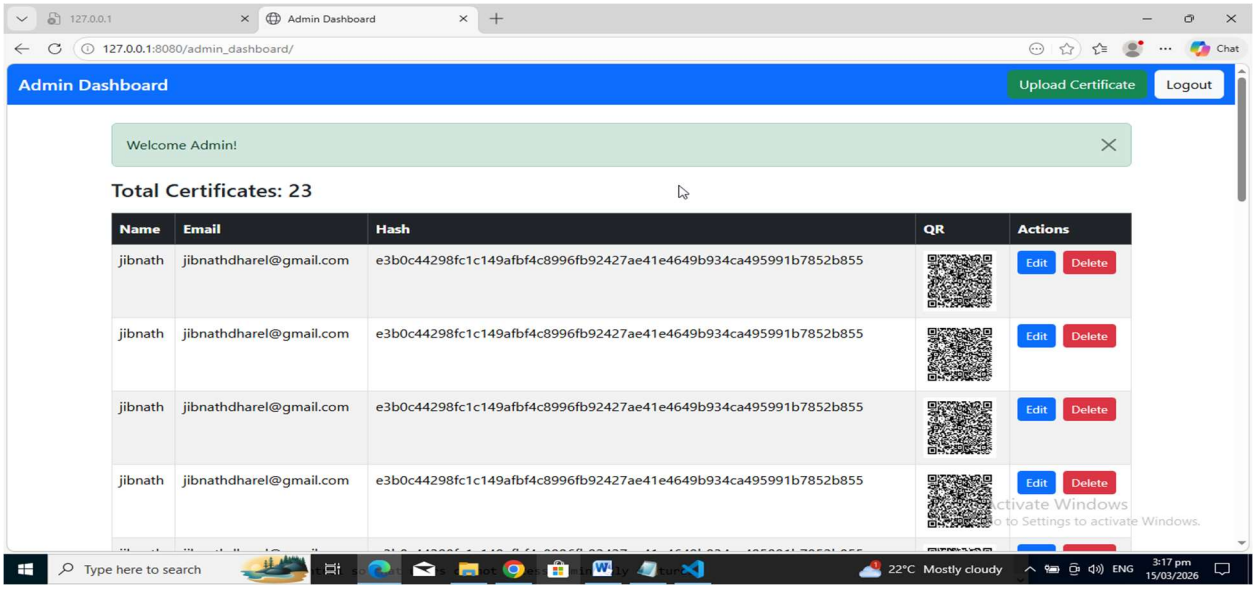

Figure 2: Admin Dashboard

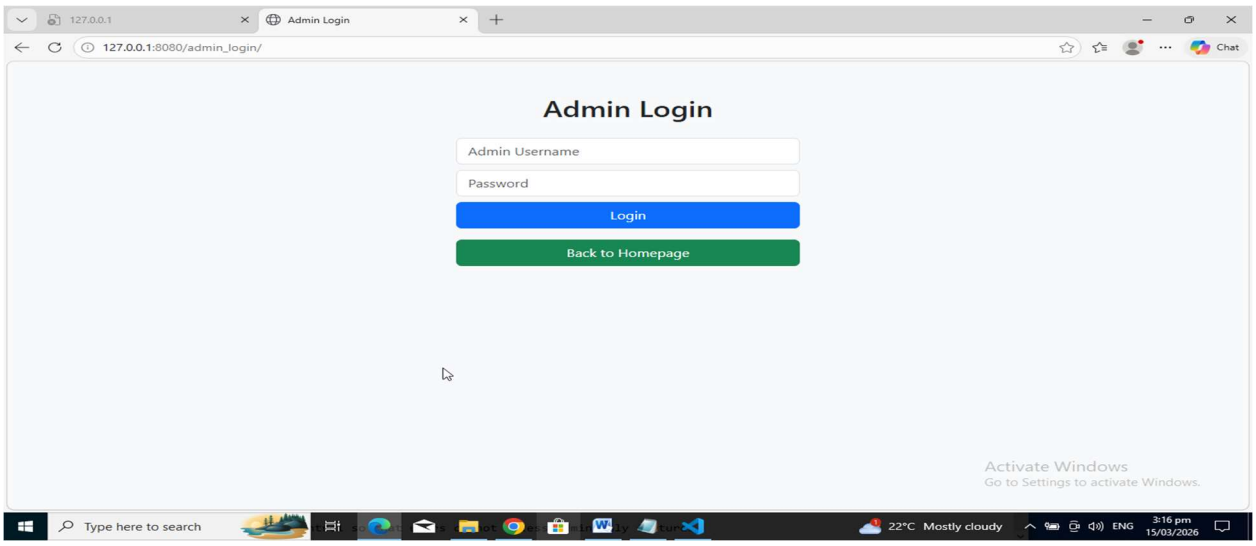

Figure 3: Admin Login

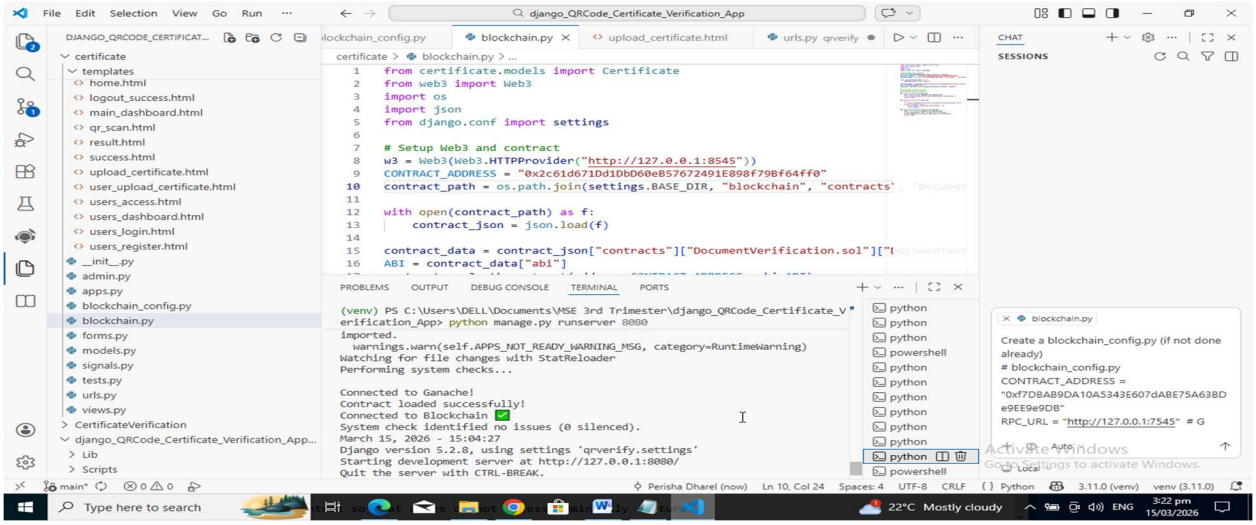

Figure 4: Ganache Running and Connected

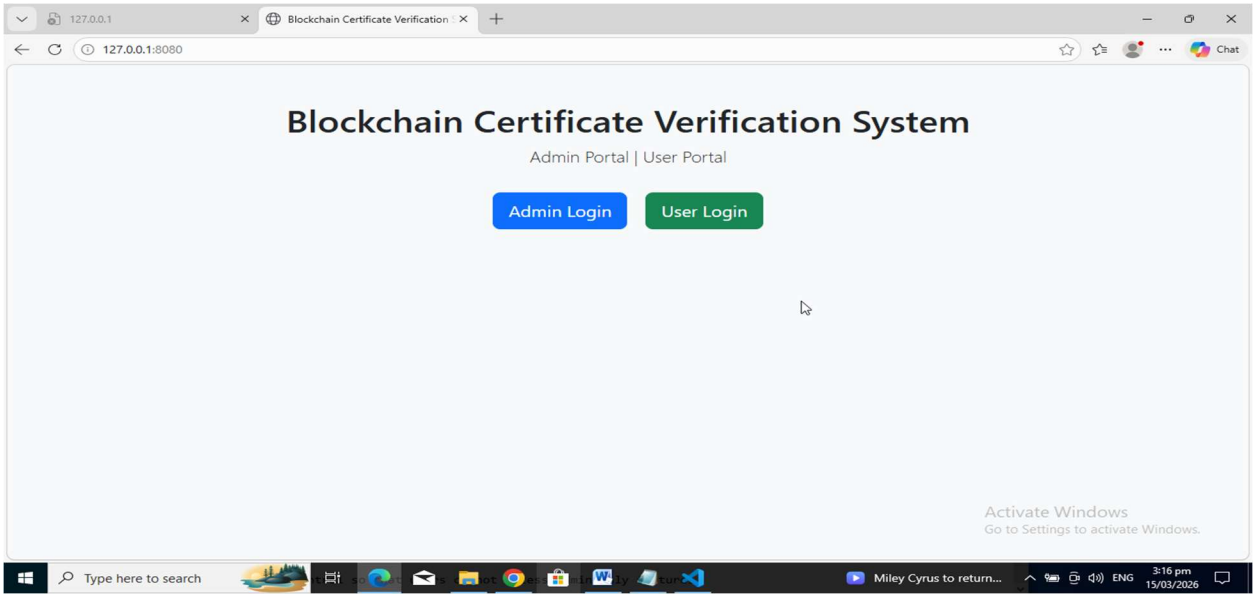

Figure 5: Home Page

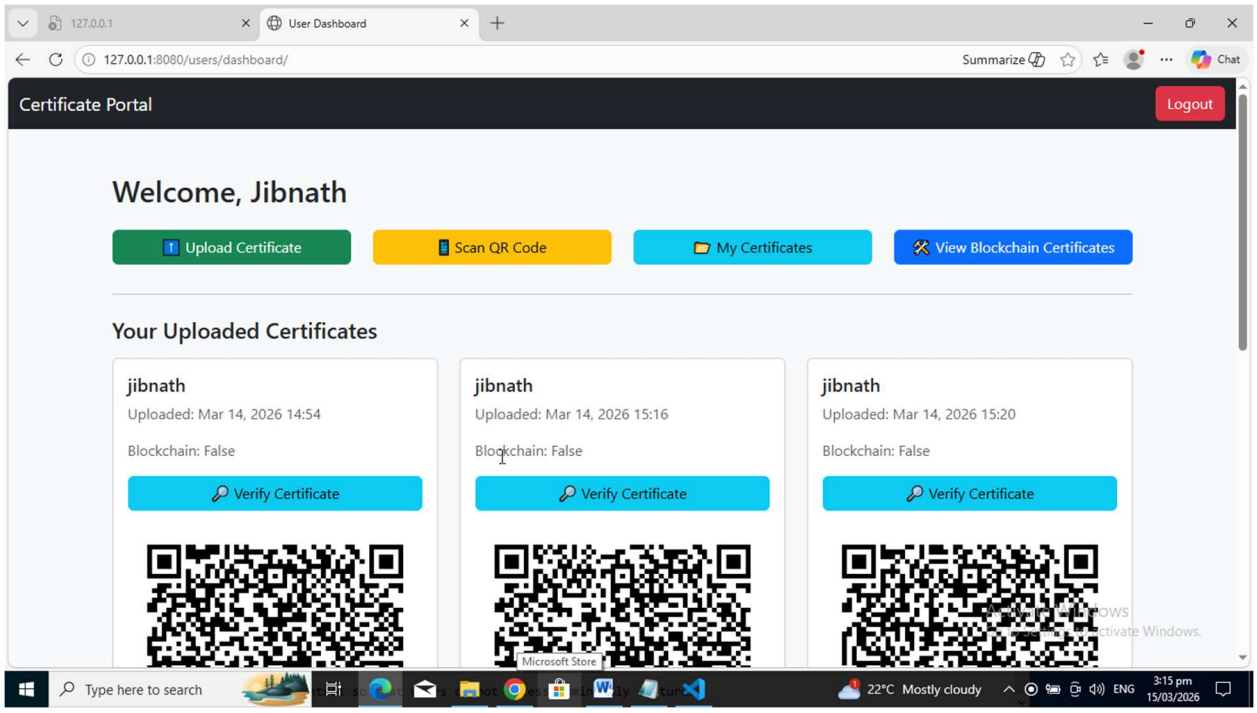

Figure 6: Users Uploaded Certificate

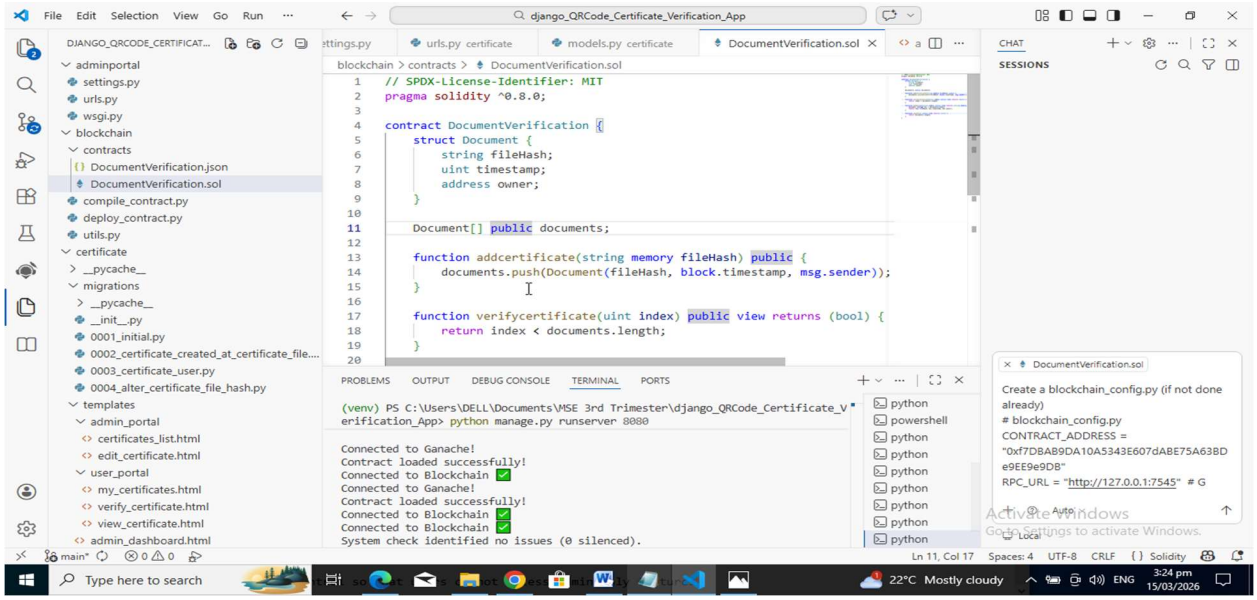

Figure 7: Solidity Contract

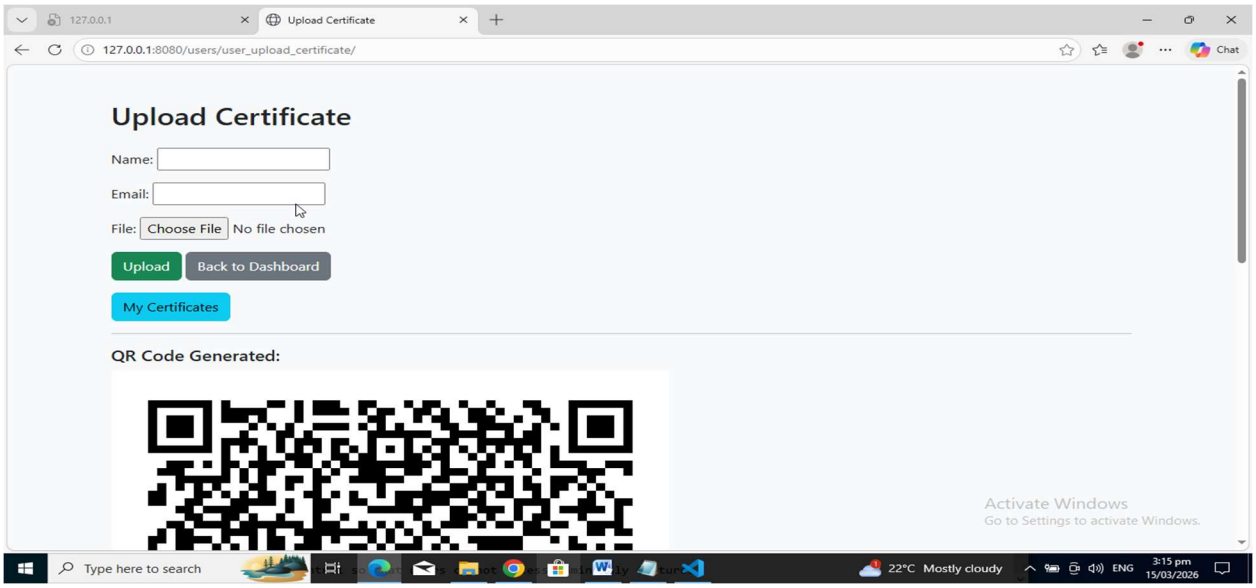

Figure 8: Upload Certificate from User

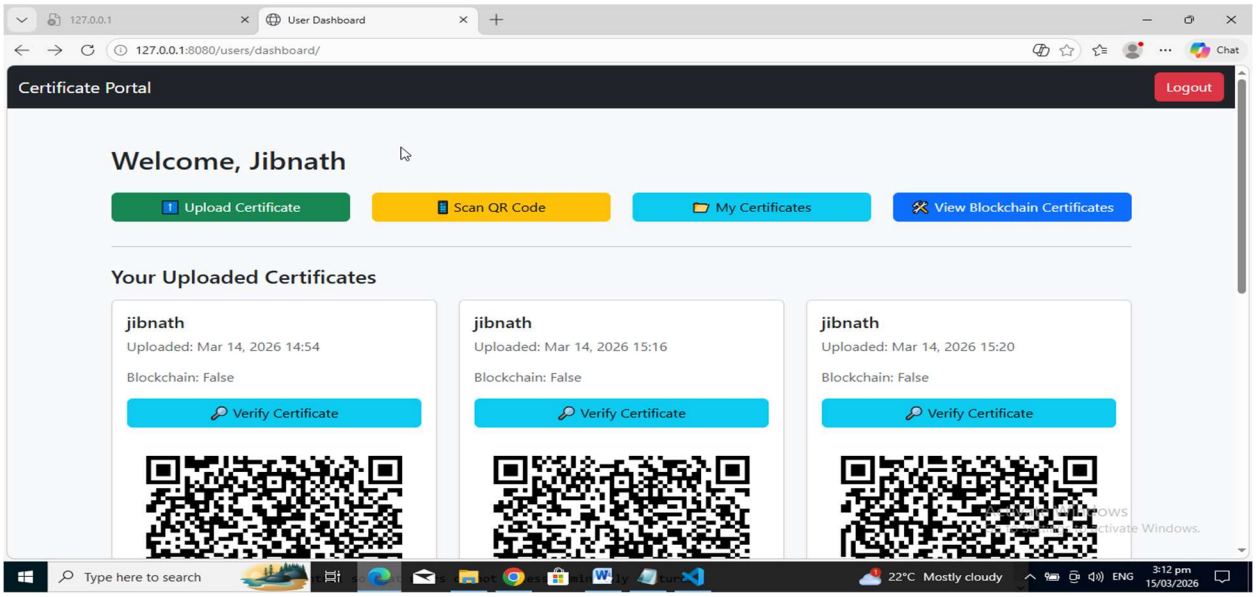

Figure 9: Uploaded Certificate of User